Design Proposal: Spike Investigation - Anomaly Detection & Attribution

Scope: deep on two of the three components - **anomaly detection** and **attribution**. Forecasting is de-scoped: it isn't in the brief's evaluation criteria, and a shallow third pass would dilute the two that matter. All numbers come from `output/` after `python run_pipeline.py`.

1. Problem framing

Two distinct ML problems, chained, not merged into one model:

1. **Anomaly detection** - unsupervised time-series outlier detection. No reliable label exists at deployment time, only a business definition of "meaningful," so the output must be a calibrated departure-from-expectation score, not a classifier.
2. **Attribution** - weakly supervised multi-label classification with unverifiable ground truth. "Why" is never observed, only inferred, and causes co-occur, so a single-label classifier would misrepresent the system's confidence.

Keeping them decoupled matters: different failure modes, different consumers, and merging them hides which half is wrong when the system misfires.

2. Data strategy

`src/data_generator.py`: 8 brands x 6-8 channels (9-channel universe) x 36 months, built as a small structural model rather than noise with labels bolted on:

- **Revenue follows a diminishing-returns response curve** (`spend x elasticity(spend) x quality x demand x noise`), so a spend cut mechanically raises ROAS and a mix shift toward efficient channels mechanically raises blended ROAS - no special-cased "if artifact then boost ROAS" rule. The artifacts fall out of the economics that produces them in real media data.
- **iROAS comes from a latent quality factor independent of spend level**, so it doesn't move for spend/mix artifacts. That asymmetry is the single strongest attribution feature. It isn't perfectly clean either - deliberately confounded by demand shocks, mirroring real incrementality work.
- **RROI** is an EWMA shrinkage of the attributable-sales ratio toward its trailing 3-month average - "stickier" than ROAS, so a single-month move RROI doesn't confirm is a tell.
- **Leading indicators** (CTR, CPC, PICR, impression share) track the same quality factor as iROAS, not spend/mix - what tells genuine gains apart from budget artifacts.
- **Six cause types** injected as latent-factor shocks, sudden (1mo) or gradual (3mo ramp), often co-occurring with a Dirichlet-split weight, calibrated to be investigation-worthy but not trivially separable. Injection rates balance *independent events per cause*, not rows - event count is what the evaluation can resolve.
- **Realism knobs:** asymmetric per-brand seasonality, ~2-3% missing periods, one brand launching mid-panel (cold-start case), CPC undefined for TV/OOH (bought on reach). Ground truth is stored **only** for evaluation, never as a feature.

Limitations: one draw from one generative model, encoding our assumptions about response curves, seasonality, and confounding. Survivorship bias is the least faithful mechanism (a reported-metric multiplier, not true placement dropout). Label density (~19%) is higher than a real book of business, to get enough events per cause to evaluate at all.

3. Model design

Anomaly detection. Not mean +/- 2 sigma. Baseline = own trailing median (6mo) + a pooled channel seasonal index, falling back to the same-channel peer trend for brand-channels with fewer than 3 own observations. `sudden_z` (1-month) and `gradual_z` (3-month trailing mean, credited only when the months agree in sign - a lightweight CUSUM) run in parallel off the same residual. Threshold selection minimizes `FP_count + 5 x FN_count` (raw counts - normalizing by class size pushes the optimum to a degenerate corner); 5:1 reflects that a missed spike leaves the manual process as the only backstop, while a false positive costs minutes. Current calibration: **precision 0.36 / recall 0.66** - the full curve (`output/detector_pr_curve.png`) is the deliverable, not this one point.

The detector is **one-sided** (`direction="up"`). Every cause in the taxonomy raises the headline metric, so scoring on `|z|` spent half the review budget on *collapses* no cause could explain, and produced incoherent narratives ("spend fell … consistent with a budget cut" attached to a ROAS drop). Gating on signed departure dominates two-sided on both axes: precision 0.36 vs 0.31, recall 0.66 vs 0.51. Collapses route to the separate performance-decline review.

Rejected: rolling mean/std (not robust to the anomaly pulling its own window); per-series STL (most series too short - as low as 10 months post-launch); a global IsolationForest (loses channel seasonality, nothing an analyst can check against the chart).

Attribution. Independent LightGBM binary classifiers, one per cause (not one softmax) - causes co-occur, and forcing probabilities to sum to 1 would misrepresent "70% likely this AND 55% likely that" as "60/40." Models are small and heavily regularized (60 trees, 7 leaves) - with 7-14 events per cause, anything larger fits noise. **Rejected:** multi-class softmax (contradicts "distribution, not a label"); rule-based attribution (brittle, no calibrated confidence); a neural multi-label model (unjustifiable at this label volume).

4. Feature engineering

All features are z-scores/ratios against the same baseline machinery used for detection, re-used for `iroas`, `rroi`, `spend`, `ctr`, `cpc`, `picr`, `impression_share`, and channel spend share:

Feature	Separates
`roas_iroas_divergence`, `roas_rroi_divergence`	genuine gain (both move) vs. artifact (ROAS only); transient vs. sustained
`spend_z`, `mix_share_z`, `brand_spend_z`, `spend_vs_brand`	budget *reallocated* (brand total flat) vs. *removed* (total down)
`ctr_z`, `picr_z`, `cpc_z` (sign-flipped)	genuine efficiency gain / creative refresh
`impr_share_z`	survivorship bias (drops with the metric spike)
`brand_coincidence`, `market_coincidence`	external demand vs. brand-specific
`shape_sudden`	creative refresh/survivorship (sudden) vs. gain/mix shift (gradual)

Lag structure: features are contemporaneous with the flagged month - this diagnoses an already-detected spike, not a future one. The structural lag matters more: **every** quantity scoring month t uses only data strictly before t, the pooled seasonal index and peer sigma included. That is enforced, not asserted - `src/test_causality.py` re-scores a truncated panel and requires bit-identical results. An earlier version failed it: pooling those two over the full panel shifted 66% of month-t scores (max 7.3 z, against a ~1 z threshold); fixing it lowered the reported numbers and made them deployable.

5. Evaluation

Attribution, without clean labels in the real world. Cross-validation is grouped by *event*, not by row, using scikit-learn's `StratifiedGroupKFold`. One event spans up to 3 months x several channels, so its rows are near-duplicates and row-wise splitting puts the same shock in train and test - a leak worth up to 0.47 PR-AUC (external demand: 0.57 row-wise vs 0.10 event-wise). So **effective sample size is events, not rows**: 60 rows from 8 events is an 8-sample problem.

Out-of-fold results (`output/attribution_metrics.csv`); lift is included because PR-AUC is not comparable across causes of differing prevalence.

cause	events	PR-AUC	base	lift
Survivorship bias	12	0.92	0.052	17.8x
Creative refresh	13	0.59	0.050	11.8x
Genuine efficiency gain	14	0.63	0.076	8.3x
External demand spike	7	0.36	0.098	3.6x

cause	events	PR-AUC	base	lift
Spend reduction artifact	12	0.30	0.103	2.9x
Mix shift artifact	**8**	**0.10**	**0.111**	**0.89x - at chance**

Top-1 hit rate is 0.61. Four causes carry strong signal, spend-reduction is modest, and **mix-shift sits at chance** - reported, not smoothed over.

The training population is the serving population - flagged months, with **both** classes restricted. That symmetry is load-bearing. An intermediate version restricted only the negatives, keeping every labeled positive because a tiny label set could ill afford losing 37% of it. It looked excellent (mix-shift 2.21x, nothing at chance) and was an artifact: 37.5% of positives were unflagged vs 0% of negatives, and since `roas_sudden_z` is a feature and `flagged` is a threshold on it, "below the threshold" *is* "positive". Unflagged-ness alone scored 2.33x - the entire headline. When a model runs downstream of a filter, both classes must come from the filtered population, and a jump on your weakest class is evidence of a leak before it is evidence of a fix.

`brand_spend_z` / `spend_vs_brand` separate budget *removed* (brand total falls) from *reallocated* (flat). Over 5 CV seeds they are worth +0.37 lift on spend-reduction, beyond seed noise, and do **not** rescue mix-shift (1.00 -> 0.87) - they earn their place on the cause they were designed for, not the one first claimed for them.

The synthetic-label model is only a cold start; the production evaluation loop is analyst-agreement rate over time (see *Productionisation*).

Predicted probabilities are mostly low - at precision 0.36 most flagged months have no injected cause, and the model correctly stays unsure rather than confabulating.

6. Causal honesty

Causal by construction: the *true* incrementality factor behind iROAS - never fully observed even in the synthetic data; the system sees a measured, demand-confounded proxy. **Associational, and presented as such:** ROAS, RROI, and every attribution output. The model is a differential-diagnosis tool, not a causal-inference engine - "consistent with a creative refresh," never "the refresh caused this," a distinction that lives in the narrative templates (`src/attribution.py`), not just here. **For a non-technical audience:** every output pairs a probability with its evidence - "creative refresh (28% likely): CTR/PICR jumped sharply" - and says "flagged, unexplained" when nothing is confident. A real causal claim needs a designed experiment (geo holdout, PSA test) or a validated MMM - neither runs here.

7. Productionisation

Retraining: monthly, on a rolling 24-36 month window. **Monitoring:** analyst agreement rate on attribution (the real accuracy proxy); feature drift on leading indicators (PSI/KS, monthly); alert-volume drift; Brier and lift-over-base per cause, so a cause at chance surfaces instead of looking authoritative - this is what keeps mix-shift's 0.89x visible rather than buried in an average. **Cold start:** exercised here (one brand launches mid-panel) - new brand-channels borrow the peer trend and sigma until a 3-month own-history floor. Next step, not built: cluster brands into spend/mix archetypes so new brands borrow from the nearest archetype, not the full pool. **Human-in-the-loop:** every analyst confirm/correct/reject is logged and feeds the next retrain - how the synthetic-label cold start gets replaced by real signal.

8. Limits - where this breaks

The simulator's assumptions are the ceiling: if real response curves, seasonality, or confounding differ materially, both models need re-validation first. **Mix-shift attribution does not work** (0.89x lift, at chance, on 8 events) and spend-reduction is modest at 2.9x - the system separates budget mechanics from genuine gains but cannot identify reallocation. Short-history brands underperform on peer priors; the causal seasonal index is blind for a panel's first 12 months. Attribution has no "none of the above" - a driver outside the taxonomy (competitor stockout, macro shock) yields uniformly low probabilities, surfaced as "flagged, unexplained." Forecasting isn't built. Best next inputs: analyst-confirmed labels and placement-level spend/revenue.